

Database Fundamentals (BCL1223) Final Exam

Answers

Question 1: SQL Creation and Queries

a) Write a creation script for all THREE (3) tables.

```
CREATE TABLE SALESPERSON (  
    SalespersonNo CHAR(4) PRIMARY KEY,  
    SalespersonName VARCHAR(15),  
    Commission NUMBER,  
    hireDate DATE  
);  
  
CREATE TABLE PRODUCT (  
    productNo CHAR(6) PRIMARY KEY,  
    productName VARCHAR(20),  
    unitPrice NUMBER(8,2)  
);  
  
CREATE TABLE SALES (  
    salespPersonNo CHAR(4),  
    productNo CHAR(6),  
    quantity NUMBER,  
    PRIMARY KEY (salespPersonNo, productNo),  
    FOREIGN KEY (salespPersonNo) REFERENCES SALESPERSON(SalespersonNo),  
    FOREIGN KEY (productNo) REFERENCES PRODUCT(productNo)  
);
```

b) Write a SQL statement to display all salespersons sorted by hire date.

```
SELECT * FROM SALESPERSON ORDER BY hireDate;
```

c) Write a SQL statement to display all salespersons with a commission more than 10%.

```
SELECT * FROM SALESPERSON WHERE Commission > 10;
```

d) Write a SQL command to display all salespersons, product number and quantity for products wrench or hammer, sort the result by salesperson.

```
SELECT s.SalespersonName, sa.productNo, sa.quantity
FROM SALESPERSON s
JOIN SALES sa ON s.SalespersonNo = sa.salespPersonNo
JOIN PRODUCT p ON sa.productNo = p.productNo
WHERE p.productName IN ('wrench', 'hammer')
ORDER BY s.SalespersonName;
```

e) Write a SQL command to display salesperson number, salesperson name, product number and quantity sold.

```
SELECT s.SalespersonNo, s.SalespersonName, sa.productNo, sa.quantity
FROM SALESPERSON s
JOIN SALES sa ON s.SalespersonNo = sa.salespPersonNo;
```

Question 2: Database Design and Concepts

a) Describe any THREE (3) basic features of a relational database.

1. **Data is organized into tables (relations):** Data is stored in two-dimensional tables consisting of rows (tuples) and columns (attributes).
2. **Primary Keys:** Each table has a primary key that uniquely identifies each record (row) in the table, ensuring data integrity and uniqueness.
3. **Foreign Keys and Relationships:** Tables can be linked to one another using foreign keys, which establish relationships between different data entities and help maintain referential integrity.

b) Create an Entity Relationship Diagram (ERD) for the given scenario.

Entities and Attributes: - STUDENT (StudentID (PK), StudentName, Address) - COURSE (CourseID (PK), CourseName, Credit) - LECTURER (LecturerID (PK), LecturerName, Phone)

Relationships and Cardinalities: - A LECTURER teaches one or many COURSE s (1:M). - A STUDENT enrolls in one or many COURSE s, and a COURSE can have many STUDENT s (M:N).

Junction Table: - ENROLLMENT (StudentID (FK), CourseID (FK), Grade) — with a composite primary key of (StudentID, CourseID).

(Note: As a text-based AI, I have provided the textual description required to draw the ERD using any diagramming tool).

Question 3: Normalization and Anomalies

a) Normalize the Production Request Form from Unnormalized Form (UNF) to Third Normal Form (3NF).

Let's assume the UNF attributes are: RequestNo, RequestDate, StaffNo, StaffName, ProductNo, ProductName, QuantityRequired.

First Normal Form (1NF): Remove repeating groups (assuming a single request can contain multiple products). We introduce a composite key (RequestNo, ProductNo).

ProductionRequest (RequestNo, RequestDate, StaffNo, StaffName, ProductNo, ProductName, QuantityRequired) *(All attributes are now atomic and the primary key is (RequestNo, ProductNo))*

Second Normal Form (2NF): Remove partial dependencies. - StaffName depends only on StaffNo. - ProductName depends only on ProductNo. - RequestDate depends only on RequestNo.

Tables in 2NF: 1. STAFF (StaffNo, StaffName) 2. PRODUCT (ProductNo, ProductName) 3. REQUEST (RequestNo, RequestDate, StaffNo) — StaffNo is a FK to STAFF. 4. REQUEST_DETAILS (RequestNo, ProductNo, QuantityRequired) — RequestNo and ProductNo are composite FKs.

Third Normal Form (3NF): Remove transitive dependencies. In this scenario, there are no apparent transitive dependencies remaining after reaching 2NF, so the design remains the same.

1. STAFF (StaffNo (PK), StaffName) 2. PRODUCT (ProductNo (PK), ProductName) 3. REQUEST (RequestNo (PK), RequestDate, StaffNo (FK)) 4. REQUEST_DETAILS (RequestNo (PK/FK), ProductNo (PK/FK), QuantityRequired)

b) Describe THREE (3) database anomalies and provide examples for each.

1. **Insertion Anomaly:** Occurs when certain data cannot be inserted into the database without the presence of other data. *Example:* In a poorly designed student-course table, you cannot add a new course until at least one student enrolls in it.
2. **Update Anomaly:** Occurs when redundant data requires updating in multiple places, and failing to update all instances leads to inconsistency. *Example:* If a lecturer's phone number is stored in every course record they teach, changing the phone number requires updating multiple records. If only one is updated, the database contains conflicting information.
3. **Deletion Anomaly:** Occurs when the deletion of a piece of data inadvertently causes the loss of other unrelated data. *Example:* If you delete the only student enrolled in a specific course from a combined table, you might lose the information about the course itself (like its credit hours) if it's not stored elsewhere.